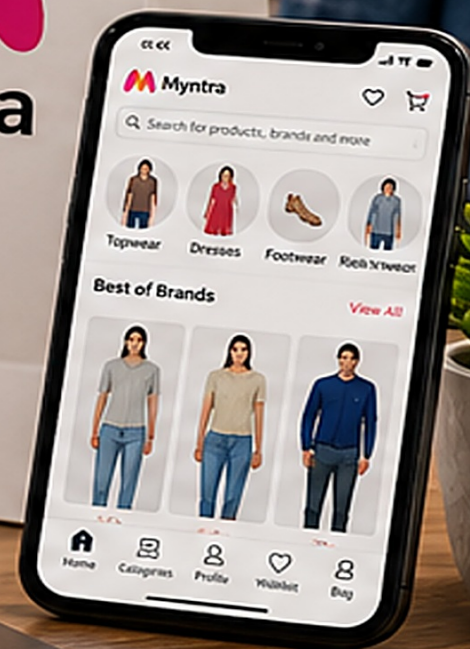




FASHION & CLOTHING ANALYTICS

EXCEL DATA ANALYSIS PROJECT

Transforming raw apparel data into actionable business insights using Excel



DATASET

Myntra Apparel Dataset
500K+ Records



TOOLS USED

Microsoft Excel
(Formulas, Functions,
Lookups)



KEY ANALYSIS

Discounts, Sizes,
Categories, Brands,
Pricing & more



OUTCOMES

Actionable Insights
to drive better
business decisions

- ✓ Data Cleaning
- ✓ Standardization
- ✓ Analysis
- ✓ Insights
- ✓ Recommendations

Analytical Framework & Project Scope

Phase A - Foundation (Data Cleaning)

- ✓ Duplicate record elimination
- ✓ Discount offer string standardization
- ✓ Algorithmic null-value handling (Prices)
- ✓ Categorical null replacement (Sizes)

Phase B - Insight Generation (Data Analysis)

- ✓ Premium product pricing trends (>4 ratings)
- ✓ Deep discount inventory volume (>50%)
- ✓ Core sizing availability (Size M)
- ✓ Dynamic discount categorization logic

Phase C - Agility (Data Retrieval)

- ✓ Array-based multi-column lookup (Product 11226634)
- ✓ Precision exact-match pricing (Product 6744434)
- ✓ Dynamic nested two-way matrix lookup

A.1 Data Cleaning: Duplicate Removal

Task

Check for duplicate values across the complete dataset and remove them to ensure baseline data integrity.

Methodology

Utilized Excel's built-in Remove Duplicates command targeting all functional columns simultaneously.

KPI

**0 Duplicates
Found**

The screenshot shows the Microsoft Excel interface with the 'Data' tab selected. A 'Remove Duplicates' dialog box is open, displaying the following options:

- ☒ Select All
- ☐ Unselect All
- ☒ My data has headers

The 'Columns' list includes:

- ☒ URL
- ☒ Product_Id
- ☒ BrandName
- ☒ Category
- ☒ Individual_category

The background data table is as follows:

URL	Product_Id	BrandName	Category	Individual_category
https://www	2290012	Roadster	Men	Men
https://www	13780125	LOCOMOT	Men	Men
https://www	11899556	Readster	Men	Men
https://www	4333679	Zivame	Men	Men
https://www	11990882	Readster	Men	Men
https://www	2409050	Mast & H	Men	Men
https://www	6744434	HIGHLAN	Men	Men
https://www	8439415	Mayra	Men	Men
https://www	17381394	Readster	Men	Men
https://www	2255257	HERE&NO	Men	Men
https://www	7695755	HRX by Hr	Men	Men
https://www	10307375	Readster	Men	Men
https://www	12874324	Anubhute	Men	Men
https://www	11634538	Athena	Men	Men
https://www	2312181	Readster	Men	Men
https://www	13342286	HIGHLAN	Men	Men
https://www	10473230	Vishudh	Men	Men
https://www	10565132	Sangria	Men	Men
https://www	12391750	Tokyo Talkies	Men	Men
https://www	2522998	Dressberry	Men	Men
https://www	17855142	Readstar	Men	Men
https://www	12133330	Anouk	Men	Men
https://www	1864573	Enamor	Men	Men
https://www	2955763	Mithex Hanva	Men	Men

A confirmation message from Microsoft Excel states: "No duplicate values found." with an OK button.

A.2 Data Cleaning: Discount Offer Standardization

Objective

Standardize highly varied text strings in DiscountOffer into uniform numerical values.

Code Block

```
=ROUND(IF([@DiscountOffer]="","0",  
SUBSTITUTE(IFERROR(SUBSTITUTE(SUBSTITUTE(  
[@DiscountOffer],"Hurry*",""),"OFF","")  
*[@OriginalPrice (in Rs)]  
SUBSTITUTE(SUBSTITUTE([@DiscountOffer],  
Hurry*",""),"OFF",""),"Rs. ", "")),2)
```

Insight

Applied nested SUBSTITUTE and IFERROR functions to systematically strip text anomalies ('Hurry*', 'OFF', 'Rs.') and output clean numeric discount values.

Product	Original Price (in Rs)	Discount Offer	Standardized Discount Offer (in Rs)
Men roadster w	825	1099 55% OFF	28,30,33,24
Men locomotive	517	1149 35% OFF	5,55,1,34
Men roadster ee	620	1205 20% OFF	30,40,6,44
Women alame wo	893	1255 35% OFF	5,16,1,55,581
Women roadster women white solid x neck	599	25% OFF	35,5,55,1,AL
Women mast harterurwomen yellow cood	599	20% OFF	X5,5,55,1,3L
Men highlander	390	1490 50% OFF	30,30,53,44
Women maya pink embroidered a line pur	1058	55% OFF	5,15,1,34
Women roadster women pack of 2 old ad	1086		X5,6,55,1,3L
Men herenon men blood him M mid ros	2740		28,30,33,6L
Men huy by bolt	1214	2699 55% OFF	5,35,1,23,441
Men midster men navy blue white obg	699		X3,5,55,1,3L
Women emulhotes	1019	2099 30% OFF	5,16,1,43,551
Women alkon women black sold beds us	2459	50% OFF	X5,5,54,1,XL
Women nactler women maroon sold rux	790	60% OFF	X3,4,54,1,XL
Men highlander	516	1099 50% OFF	39,40,43,44
Women sishad or	606	1099 52% OFF	5,15,1,16,831
Women sengre women green off ulicy pr	3099		X3,5,55,1,3L
Women drestharry women lordy oloo typs	2149	61% OFF	20,20,32,24
Women roadster women park of 8 purker	1595	55% OFF	X3,5,54,1,XL

A.3 Data Cleaning: Targeted Average Imputation

Task

Identify rows where both DiscountPrice and DiscountOffer are null, filling the price with the exact average of the product's specific category.

```
=IF(AND([@[DiscountPrice (in Rs)]]="",[@[DiscountOffer]]=""),  
AVERAGEIF([Category],[@Category],[DiscountPrice (in Rs)]),  
[@[DiscountPrice (in Rs)]])
```

Strategic Value

Maintained category-level pricing accuracy. By conditionally applying AVERAGEIF, high-end categories don't skew the missing values of budget categories.

	I	J	K	L	M	N	O	P	Q	R	S	T
1	OriginalPrice (in Rs)	DiscountOffer	SizeOption	Ratings	Reviews	Standardized_DiscountOffer (in Rs)	DiscountPrice_Filled					
2	1499	45% OFF	28, 30, 32, 34, 36	3.9	999	₹ 674.55	=IF(AND([@[DiscountPrice (in Rs)]]="",[@[DiscountOffer]]=""),					
3	1149	55% OFF	S, M, L, XL	4	999	₹ 631.95	AVERAGEIF([Category],[@Category],[DiscountPrice (in Rs)]),					
4	1399	55% OFF	38, 40, 42, 44, 46, 46	4.3	999	₹ 769.45	[@[DiscountPrice (in Rs)])]					
5	1295	31% OFF	S, M, L, XL, XXL	4.2	999	₹ 401.45	₹ 893.00					
6	599	35% OFF	X5, S, M, L, XL	4.2	999	₹ 239.65	₹ 0.00					
7	599	40% OFF	X5, S, M, L, XL	4.4	998	₹ 239.60	₹ 0.00					
8	1499	60% OFF	30, 32, 34, 36	3.5	998	₹ 899.40	₹ 599.00					
9	1395	55% OFF	S, M, L, XL	3.7	998	₹ 809.10	₹ 0.00					
10	1098		X5, S, M, L, XL	4.3	997	₹ 0.00	₹ 1,790.09					
11	2749		28, 30, 32, 34, 36	3.5	996	₹ 0.00	₹ 1,351.21					
12	2699	55% OFF	S, M, L, XL, XXL	4.4	996	₹ 1,484.45	₹ 1,214.00					
13	699		X5, S, M, L, XL, 3XL, 3XL, 4XL	4.1	996	₹ 0.00	₹ 1,138.63					
14	3399	70% OFF	S, M, L, XL, XXL	4.3	996	₹ 2,379.30	₹ 1,019.00					

Myntra Fasion Clothing

A.4 Data Cleaning: Size Availability Status

Task

Replace all null values in the SizeOption column with the text 'Not Available'.

Methodology

Executed a standard, full-sheet Find & Replace operation.

Insight

The dataset was already fully populated in the categorical size dimension prior to intervention.

**No Null Values
Found**

Microsoft Excel



We couldn't find anything to replace. Click Options for more ways to search.

FYI: It's possible the data you're trying to replace is in a protected sheet. Excel can't replace data in protected sheets.

OK

B.1 Data Analysis: Premium Pricing Trends

Business Query

Calculate the overall average original price for highly-rated products (Ratings > 4).

₹ 1,966.67

```
=AVERAGEIF(Table1[Ratings], ">4", Table1[Original Price (in Rs)])
```

The screenshot shows an Excel spreadsheet with a table of product data. The formula bar displays the formula: `=AVERAGEIF(Table1[Ratings], ">4", Table1[Original Price (in Rs)])`. The table has columns for Product ID, Product Name, Size, Color, Rating, and Price. The data is filtered to show products with a rating greater than 4. The average original price for these products is calculated as ₹ 1,966.67.

Product ID	Product Name	Size	Color	Rating	Price
1089	S2H OFF	28, 30, 32, 34, 36		2.0	990
1149	65H OFF	3, 5A, L, XL		4	999
1089	50H OFF	28, 40, 44, 46, 48, 50		4.2	999
1095	35H OFF	3, 9S, L, XL, 4XL		4.2	999
599	50H OFF	32, 34, 36, 38, 40		4.2	999
599	30H OFF	30, 32, 34, 36, 38		4.2	999
1499	50H OFF	30, 32, 34, 36, 38		3.7	999
1259	50H OFF	3, 5S, L, XL		3.7	996
1096		39, 5, 33, L, 4XL		4.3	997
2740		38, 50, 52, 25, 35		3.5	956
2829	32H OFF	3, 20, 1, M, 2XL		4.4	956
699		33, 5, 33, L, 5L, 5XL, 6XL		4.1	956
3559	10H OFF	3, 5L, L, H, 4XL		4.3	956
2469	60H OFF	5S, 0, 5A, L, 3XL		4.3	956
759	00H OFF	36, 5, 30, L, 4L		4	955
2099	50H OFF	30, 34, 36, 38, 40		4.2	955
1056	38H OFF	3, 5S, L, 4S, 6S		4.2	955
3960		36, 5, 5S, L, 5L, 5XL, 59L		4.3	955
2149	21H OFF	26, 28, 30, 32, 34		4.1	955
599	75H OFF	30, 32, 34, 36, 38		4.1	955
1296		39, 5, 30, L, 4L		4.4	954
1999	55H OFF	36, 5, 36, L, 3L, 3XL		3.5	953
859		3, 5S, L, M		4.2	993

B.2 Data Analysis: Deep Discount Inventory

Business Query

Quantify the absolute volume of products carrying a discount offer greater than 50% OFF.

Syntax:

```
=COUNTIF(Table1[Standardized_DiscountOffer  
(in %)], ">50%")
```

221,864 Products

Insight Note: Over 220,000 SKUs are currently operating at a steep discount, representing a major driver of potential conversion volume.

The screenshot shows the Microsoft Excel application window. The title bar indicates the file is named 'Myntje Fashion Oct... - Saving...'. The ribbon is set to the 'Formulas' tab, showing various formula-related icons. The spreadsheet itself is titled 'Regalpa Fashion Clothing' and contains data organized in columns. The columns are labeled: 'DowColours', 'Instage', 'Benamen', 'Standaard, DoonaeDffer En IL', 'Standaard, fline', and 'Standaard, DoonaeDffer En IL'. The data is organized in rows, with some cells containing numerical values and others containing text descriptions of clothing items. The Excel ribbon is visible at the top, and the Windows taskbar is at the bottom.

The screenshot shows the Microsoft Excel application window. The title bar reads "Myrta Fusion Cloud - Saved". The ribbon is set to "Formulas", and the "Countifs" function is selected. The formula bar contains the formula: `=COUNTIF(Table1[(Standardized_DiscountOffer (in %))>500])`. The spreadsheet displays a table with columns labeled I, J, K, L, M, N, O, P, Q, R, S. The data includes columns for "Discount", "Standardized_DiscountOffer (in %)", and "Standardized_DiscountOffer (in %)". The status bar at the bottom indicates "Myrta Fusion Cloud - Saved" and "Myrta Fusion Cloud - Saved".

B.3 Data Analysis: Core Sizing Availability

Business Query

Count the exact number of products available in size 'M'.

Syntax:

```
=COUNTIF(Table1[SizeOption], "*M*")
```

S, M, L, XL

Technical Callout: Utilized wildcard string matching (*M) to successfully isolate and extract target sizes from mixed, comma-separated size arrays.

656 Products

SizeOption	Ratings	Reviews	Standardised_DiscountOffer (Rs)	DiscountPrice_Rfled	Standardised_DiscountOffer (%)
28,30,32,34,36	2.0	990	1,673.25	1,624.00	49%
S, M, L, XL	4	990	1,631.92	1,527.00	55%
38,40,42,44,46,48	4.2	999	1,769.42	1,629.00	53%
S, M, L, XL	4.2	999	1,501.45	1,693.00	31%
36, S, M, L, XL	4.2	999	1,200.02	1,000.00	33%
35, S, M, L, XL	4.4	950	1,235.00	1,000.00	48%
30, 34, 36, 38	3.5	998	1,654.00	1,593.00	60%
S, M, L, XL	3.7	998	1,609.10	1,000.00	58%
28, S, M, L, XL	4.5	997	1,000.00	1,179.00	0%
28, 30, 32, 34, 36	3.5	996	1,000.00	1,194.25	0%
S, M, L, XL	3.1	996	1,104.42	1,114.00	55%
34, S, M, L, XL, 36, 38, 40, 42, 44, 46, 48	4.1	996	1,600.00	1,139.00	0%
S, M, L, XL	4.2	996	1,239.20	1,109.00	20%
35, S, M, L, XL	4.3	996	1,249.60	1,000.00	60%
36, S, M, L, XL	4	995	1,632.40	1,000.00	0%
26, 30, 34, 38	4.1	955	1,582.47	1,316.00	53%
S, M, L, XL	4.2	995	1,102.41	1,090.00	90%
34, S, M, L, XL, 36, 38, 40, 42, 44, 46, 48	4.3	953	1,000.00	1,023.00	0%
38, 30, 32, 34	4.1	995	1,110.08	1,000.00	67%
34, S, M, L, XL	4.3	955	1,742.23	1,000.00	75%
35, S, M, L, XL	4.4	994	1,600.00	1,179.00	0%
28, S, M, L, XL, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48	3.2	993	1,679.42	1,119.00	53%
S, M, L, XL	4.2	993	1,000.00	1,513.00	0%
38, 30, 32, 34, 36	3.9	995	1,104.42	1,388.00	53%
30, 34, 36, 38, 40, 42, 44, 46, 48	4.3	996	1,269.50	1,123.00	50%
S, M, L, XL	4.3	996	1,475.60	1,150.00	60%
36, S, M, L, XL	4.2	996	1,000.00	1,000.00	60%
26, S, M, L, XL	4.2	996	1,102.37	1,506.00	50%
38, 30, 32, 34	4	995	1,706.62	1,000.00	20%
36, S, M, L, XL	4.4	995	1,000.00	1,000.00	20%
26, S, M, L, XL	4.3	995	1,000.00	1,000.00	60%
26, S, M, L, XL, 34, 36, 38, 40, 42, 44, 46, 48	4.4	995	1,250.08	1,149.00	0%
28, 30, 32, 34	4.1	997	1,000.00	1,179.20	55%
31, M, L, XL	4.3	996	1,630.68	1,203.00	59%
36, S, M, L, XL	4.3	996	1,239.30	1,139.00	53%
38, 30, 32, 34	4.3	995	1,575.60	1,260.00	90%
34, S, M, L, XL	4.2	995	1,683.47	1,316.00	62%
34, S, M, L, XL, 36, 38, 40, 42, 44, 46, 48	4.3	995	1,103.06	1,523.00	0%
38, 30, 32, 34	4.3	994	1,243.06	1,000.00	75%
36, S, M, L, XL	4.4	994	1,000.00	1,179.00	0%
28, S, M, L, XL, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48	3.6	993	1,679.65	1,119.00	55%
38, 30, 32, 34	2.1	893	1,114.1	1,611.14	0%

B.4 Data Analysis: Dynamic Categorization

Business Query

Automate product labeling based on discount thresholds.

Syntax:

```
=IF([@[Standardized_DiscountOffer  
(in %)]>50%,"High Discount","Low  
Discount")
```

Discount
Value

High Discount
(if > 50%)

Low Discount
(if < 50%)

Strategic Outcome

Enables marketing and sales teams to instantly filter the dataset by promotional aggression without manual calculation.

The screenshot shows a WPS Spreadsheet interface. The formula bar displays the formula: `=if([@[Standardized_DiscountOffer (in %)]>50%,"High Discount","Low Discount")`. The spreadsheet contains a table with the following columns: Rating, Standardized_DiscountOffer, and Discount_Label. The data is as follows:

Rating	Standardized_DiscountOffer	Discount_Label
5	990	40% High Discount
5	999	59% High Discount
5	999	35% High Discount
5	999	35% High Discount
5	999	35% Low Discount
5	999	45% High Discount
5	998	40% High Discount
5	998	50% High Discount
5	998	50% High Discount
5	997	0% High Discount
5	996	0% Low Discount
5	996	59% Low Discount
5	996	0% Low Discount
5	996	29% High Discount
5	996	70% High Discount
5	996	30% High Discount
5	996	60% High Discount
5	995	53% High Discount
5	995	55% High Discount
5	995	53% High Discount
5	995	59% High Discount
5	995	0% High Discount
5	995	63% Low Discount
5	995	50% Low Discount
5	994	79% Low Discount
5	994	0% Low Discount
5	993	55% Low Discount
5	993	0% Low Discount
5	993	55% Low Discount

C.1 Data Retrieval: Multi-Column Array VLOOKUP

Query Target: Find Brand, Price, and Rating for Product ID: 11226634

```
=VLOOKUP(11226634,Table1[#All],[Product_id]:[Standardized_DiscountOffer (in Rs)],{2,8,11},FALSE)
```

1

Brand

Maniac

2

Price

₹ 1,199

3

Rating

3.9

The screenshot shows a WPS Spreadsheet window titled 'Mynta Fashion Catalog - Spring...'. The formula bar displays the VLOOKUP formula: `=VLOOKUP(11226634,Table1[#All],[Product_id]:[Standardized_DiscountOffer (in Rs)],{2,8,11},FALSE)`. The spreadsheet contains a table with the following columns: **Product_id**, **Brand**, **Standardized_DiscountOffer (in Rs)**, **Standardized_DiscountOffer (in Rs)**, and **Standardized_DiscountOffer (in Rs)**. The table lists various products with their IDs, brands, and prices. The result of the VLOOKUP formula is displayed in cell S7, showing the brand 'Maniac', the price '1100', and the rating '3.9'.

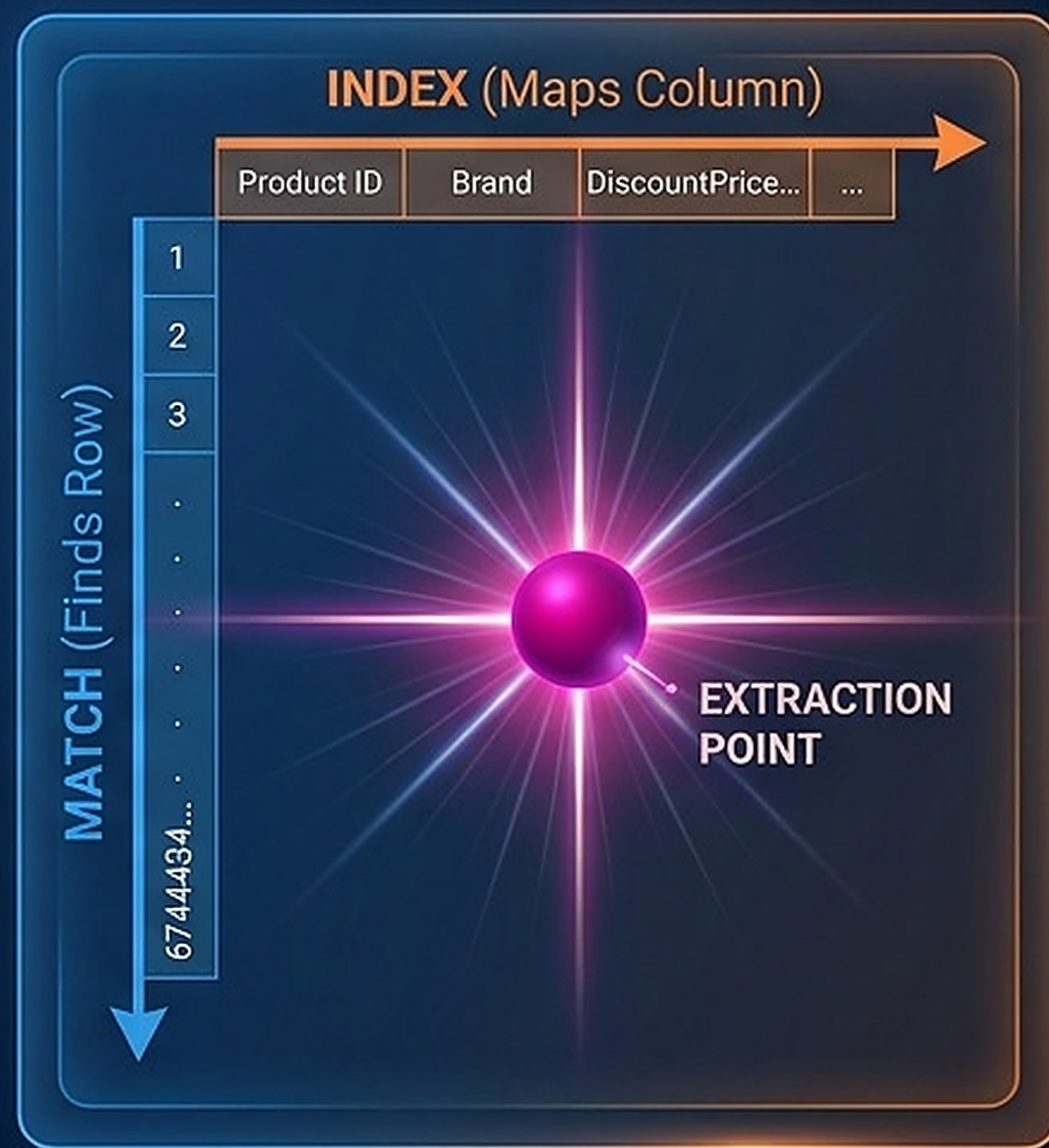
Product_id	Brand	Standardized_DiscountOffer (in Rs)	Standardized_DiscountOffer (in Rs)	Standardized_DiscountOffer (in Rs)
3.0	000	₹ 575.25	₹ 625.00	40% Discount
4	000	₹ 533.25	₹ 527.00	30% 10% Discount
6.3	000	₹ 755.45	₹ 625.00	35% 10% Discount
4.2	000	₹ 150.45	₹ 053.00	37% 10% Discount
4.3	000	₹ 125.45	₹ 0.00	30% 10% Discount
4.1	000	₹ 210.00	₹ 0.00	40% 10% Discount
3.0	000	₹ 559.20	₹ 530.00	40% 10% Discount
3.7	000	₹ 050.20	₹ 0.00	36% 10% Discount
4.5	007	₹ 0.00	₹ 1,700.00	0% 10% Discount
2.5	000	₹ 0.00	₹ 1,751.21	0% 10% Discount
0.4	000	₹ 1,005.06	₹ 2,234.00	50% 10% Discount
4.1	000	₹ 0.00	₹ 1,210.40	0% 10% Discount
4.1	000	₹ 2,310.20	₹ 1,029.00	100% 10% Discount
4.3	000	₹ 1,168.00	₹ 0.00	40% 10% Discount
4	005	₹ 319.20	₹ 0.00	50% 10% Discount
4.2	000	₹ 1,00.17	₹ 326.00	50% 10% Discount
4.2	000	₹ 1,000.34	₹ 1306.00	30% 10% Discount
6.1	000	₹ 0.00	₹ 523.00	0% 10% Discount
6.1	000	₹ 1,210.00	₹ 0.00	50% 10% Discount
4.2	004	₹ 750.25	₹ 0.00	75% 10% Discount
6.3	000	₹ 0.00	₹ 1,706.00	50% 10% Discount

C.2 Data Retrieval: Precision Matching

Query Target: Find 'DiscountPrice' for Product ID: 6744434

```
=INDEX(Table1[DiscountPrice  
(in Rs)],MATCH(6744434,Tabl  
e1[Product_id],0))
```

₹ 599



The screenshot shows a spreadsheet with a table containing product data. The columns are Product ID, Brand, DiscountPrice, and The row corresponding to Product ID 6744434 is highlighted, showing a DiscountPrice of ₹ 599.

Product ID	Brand	DiscountPrice	...
6744434		₹ 599	

C.3 Data Retrieval: Nested XLOOKUP Matrix

Query Target:

Dynamically extract OriginalPrice (in Rs) for Product ID: 2296012

```
=XLOOKUP(S8,Table1[Product_id],  
XLOOKUP(S10,Table1[#Headers],  
[BrandName]:[Discount_Label],Table  
1[BrandName]:[Discount_Label]))
```

- Creates a robust intersection lookup searching both rows and columns dynamically, eliminating brittle hard-coded column index numbers.

KPI

₹ 1,499

[illegible]

Methodological Synthesis: Data Retrieval Strategies

	VLOOKUP (Ref: C.1)	INDEX/MATCH (Ref: C.2)	XLOOKUP (Ref: C.3)
Syntax Complexity	🟡 Moderate	⚠️ High (Nested logic)	✅ Modern/Streamlined
Flexibility	🔒 Rigid (Left-to-Right only)	↔️ High (Omni-directional)	∞ Ultimate (Matrix intersections)
Ideal Use Case	Static, multi-column array extraction.	Resilient lookups on large, shifting datasets.	Dynamic, two-way cross-referencing dashboards.

Final Takeaway: Evolving beyond standard VLOOKUP mechanics to nested XLOOKUP architectures enables vastly more agile, error-resistant business intelligence dashboards.